One Model, Many Geometries: Spectral Dynamics of Multilingual LLMs

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Introduction. Multilingual training dynamics of LLMs have largely been studied using only feature-level methods (Blevins et al., 2022; Wang et al., 2024; Chang et al., 2022). However, Li et al. (2025) show that global spectral metrics can also be used to track the evolution of representation geometry during pretraining. We hypothesize that the macro-level co-evolution of languages predicts multilingual capabilities, and we preliminary investigate whether languages exhibit synchronized geometric phases. In two multilingual LLMs, we find that 14 diverse languages exhibit almost identical phases, but differ in effective dimensionality of their representations. Furthermore, we find that differences cannot be explained by amount of training data or language similarities.

Approach. Our primary metric for analyzing geometry evolution is RankMe (Garrido et al., 2023; Li et al., 2025), a measure of effective rank. For a layer L with hidden dimension D, we collect $N \gg D$ last-token activations from N distinct sequences into a matrix X. We then compute the centered covariance matrix Σ, and normalize its eigenvalues λ to yield a probability distribution $p_k = \lambda_k / \sum_j \lambda_j$. RankMe is the exponentiated entropy of p: $\exp(-\sum_k p_k \ln p_k)$.

We compute RankMe for all layers L across all checkpoints of our models.

Experiments. We use two open-source 8B multilingual LLMs: FuxiTranyu (Sun et al., 2024) (57 checkpoints) and Apertus (Apertus et al., 2025) (44 checkpoints). Since both models have $D = 4096$, we collect activations from $N = 10,000$ distinct sequences per language from `FineWiki` (Penedo, 2025)[1], filtering and truncating them to have between 32 and 256 tokens. We select 14 languages to cover diverse families and representation in training data (see A.1 for details).

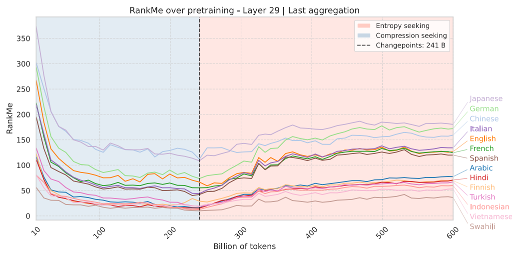

Figure 1: RankMe evolution of FuxiTranyu.

Figure 1 shows the evolution of RankMe at the last layer of FuxiTranyu, and Figure 3 shows the same for Apertus. In line with Li et al. (2025), we find distinct phases of geometry evolution in both models. Moreover, we find that all languages essentially exhibit the same phases, although with differing RankMe values.

We observe some clustering by language family (e.g. Spanish, French and Italian), but families do not explain differences in general. Training data distribution is not a good predictor either, as English – the most represented language – sits somewhere in the middle in both models. However, we do find statistically significant correlations between RankMe and tokenizer efficiency, although they do not hold for both models (see A.4). In particular, tokenizer fertility shares a significant negative correlation with RankMe in FuxiTranyu's last three layers, suggesting that, in layers dominated by syntax processing, increased fertility restricts the model's effective dimensionality.

Next Steps. We will continue investigating factors that could explain RankMe differences (e.g. tokenizer fertility, vocabulary overlap). We have collected downstream performance data, and we will run correlation analyses. Finally, we will explore layer-wise representation geometry through a spectral lens, and conduct ablation studies.

[1]After some initial experiments with `FineWeb2` (Penedo et al., 2025), we found that a clean and structurally consistent dataset like FineWiki can better highlight the differences and similarities between languages in their geometric evolution.

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A Appendix

A.1 Languages

We compute RankMe for 14 languages from diverse families and with varying level of representation in models' training data. Tables 1, 2 and 3 show the languages, the families and the distribution in training data.

Language	Family	Branch
English	Indo-European	Germanic
Chinese	Sino-Tibetan	Sinitic
German	Indo-European	Germanic
French	Indo-European	Romance
Spanish	Indo-European	Romance
Italian	Indo-European	Romance
Arabic	Afro-Asiatic	Semitic
Vietnamese	Austroasiatic	Mon-Khmer
Japanese	Japonic	Japanese
Turkish	Turkic	Oghuz
Finnish	Uralic	Finnic
Hindi	Indo-European	Indo-Iranian
Indonesian	Austronesian	Malayo-Polynesian
Swahili	Niger-Congo	Benue-Congo
Total	**9 families**	**11 branches**

Table 1: Language families and branches.

Language	FuxiTranyu 8B
English	28.80%
Chinese	10.40%
German	8.10%
French	7.30%
Spanish	5.70%
Italian	3.70%
Arabic	1.70%
Vietnamese	1.60%
Japanese	1.50%
Turkish	1.49%
Finnish	0.94%
Hindi	0.79%
Indonesian	0.33%
Swahili	0%
Total	**72.35%**

Table 2: Training data coverage by language in FuxiTranyu. Data was provided by the authors.

Language	Apertus 8B	FineWeb2
English	~60%	0%
Chinese		12.66%
German		9.36%
French		7.28%
Spanish		8.88%
Italian		4.80%
Arabic		1.26%
Vietnamese	~40%	0.89%
Japanese		8.23%
Turkish		1.94%
Finnish		0.73%
Hindi		0.45%
Indonesian		2.04%
Swahili		~0%

Table 3: Training data coverage by language in Apertus. As reported in the technical report, Apertus was trained on roughly 60% English data and 40% non-English data (Apertus et al., 2025). As we do not have access to the full training distribution, we include the language distribution in `FineWeb2` to provide in indication of multilingual data distribution.

A.2 Last-layer RankMe evolution of FuxiTranyu

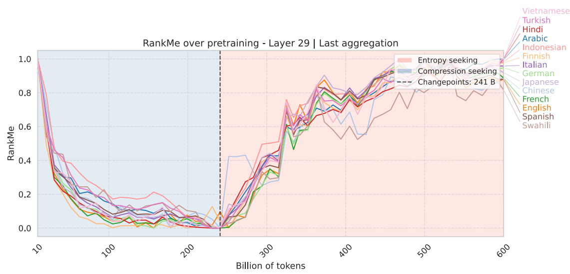

Figure 2: Normalized last-layer RankMe evolution of FuxiTranyu. Min-max scaling values per phase shows that languages exhibit the same geometric phases.

A.3 Last-layer RankMe evolution of Apertus

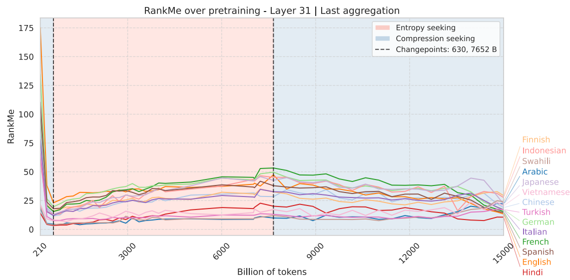

Figure 3: Last-layer RankMe evolution of Apertus. The model exhibits 3 phases, with an initial compression, an intermediate expansion, and a final compression. The higher training token count of Apertus with respect to FuxiTranyu (15T vs 600B) allows to analyze geometric co-evolution in longer training regimes.

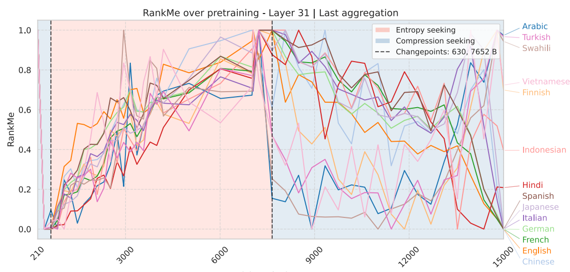

Figure 4: Normalized last-layer RankMe evolution of Apertus. Min-max scaling values per phase shows that languages exhibit the same geometric evolution in the first two phases, but differ in the last one, where compression is non-uniform and some languages exhibit a final expansion.

A.4 Correlation with tokenizer fertility

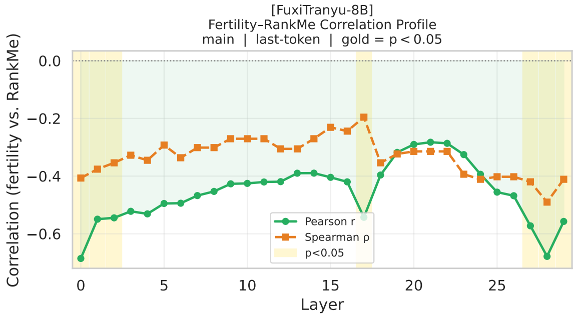

Figure 5: Pearson and Spearman correlation coefficients between tokenizer fertility and RankMe score for all languages across all layers of the final Fuxi model. Gold bands show statistically relevant correlation. We observe a statistically significant correlation on the first 3 and last 3 layers, where the representation space is still strictly linked to surface-level syntax. Overall, the correlation is negative across all layers, indicating that more fertile languages utilise on average a smaller proportion of the embedding space.

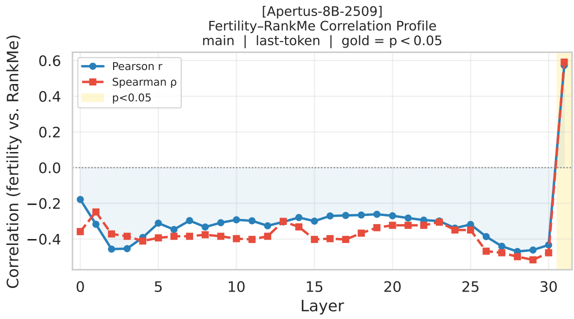

Figure 6: Pearson and Spearman correlation coefficients between tokenizer fertility and RankMe score for all languages across all layers of the final Apertus model. Gold bands show statistically relevant correlation. In contrast with Fuxi, we don't observe a statistically significant correlation in any of the layers except the last one. Contarry to Fuxi, this last correlation is positive. It is hard to derive conclusions that can justify this, especially given that the two models behave very differently.